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**ALLOMETRIC SCALING AND NEURO-SYMBOLIC
TRACKING FOR AUTOMATED PEDIATRIC CENSUS
IN CLINICAL ENVIRONMENTS**

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Declaration

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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*Dedicated to the children whose stories are often untold, and to the pursuit of
visibility in every clinical data point.*

Abstract

This research addresses the systemic “Invisible Child” phenomenon in Ghanaian pediatric departments, where cultural carrying practices (infants on mothers’ backs) lead to significant demographic undercounting. We propose a hybrid mathematical framework that synthesizes scale-invariant Allometric Scaling with YOLOv11-seg instance segmentation and Kalman-filtered DeepSORT tracking. By utilizing biological constants (Head-to-Body ratios), the system mathematically separates occluded children from caregivers in low-resolution 640×480 clinical feeds. Calibrated against the Child Detection Dataset (CDD), the model aims to minimize Mean Absolute Percentage Error (MAPE) to $< 10\%$, providing a privacy-preserving Force Multiplier for hospital administrators at KATH.

Acknowledgment

I would like to express my deepest gratitude to my supervisor, Prof. [Supervisor Name], for their invaluable guidance and technical insights. Special thanks to the staff at Komfo Anokye Teaching Hospital (KATH) for providing the clinical context that shaped this research. Finally, I thank my partner and colleagues for their support and collaboration throughout this project.

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List of Abbreviations

KATH Komfo Anokye Teaching Hospital

LHIMS Lightwave Health Information Management System

YOLO You Only Look Once

CDD Child Detection Dataset

NMS Non-Maximum Suppression

MAPE Mean Absolute Percentage Error

DeepSORT Simple Online and Realtime Tracking with a Deep Association
Metric

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CHAPTER 1

Introduction

1.1 Background

In the evolving landscape of digital healthcare, the precision of patient data is the bedrock of operational efficiency. This research explores the synthesis of biological scaling laws and computer vision to automate the pediatric census at Komfo Anokye Teaching Hospital (KATH).

1.2 Problem Statement

In major Ghanaian hospitals, optimizing medical staff allocation through the Local Health Information Management System (LHIMS) requires precise demographic data. However, the “Invisible Child” phenomenon—infants carried on mothers’ backs—creates overlapping geometries that standard algorithms actively erase as visual noise (Addotey-Delove et al., 2023).

1.3 Research Gap

Specialized models like YOLOCDD have been developed for controlled medical scenes (Diop et al., 2025), but there is a need to extend these to high-density occlusion in African clinical norms.

1.4 Research Objectives

- Formulate a spatial detection model to mathematically separate carried children from caregivers.

- Apply physical constraints via Head-to-Body Ratios to prevent misclassification.
- Implement geometric centroid tracking utilizing Kalman Filters for unique patient IDs.
- Design a localized Edge computing architecture that outputs strictly numerical matrices.

CHAPTER 2

Literature Review

2.1 Evolution of Object Detection

The transition from feature engineering (Viola & Jones, 2001) to deep learning (YOLO) has revolutionized detection, though general models often fail in clinical scenes (Diop et al., 2025).

2.2 Pediatric Anthropometry

Infant anatomy (1/4 ratio) differs significantly from adults (1/7.5 ratio), necessitating scale-invariant modeling (Mlinari et al., 2024; Lin et al., 2022).

2.3 Handling Occlusion in Ghanaian OPDs

The “Invisible Child” phenomenon is a documented failure point in standard NMS algorithms (Addotey-Delove et al., 2023).

2.4 Specialized Pediatric Datasets

The Child Detection Dataset (CDD) provides real-world clinical benchmarks for fine-tuning.

2.5 Clinical Monitoring and Accuracy

High-precision tracking is critical for infantile spasm detection and NICU monitoring (Diop et al., 2024; Keles & Bagci, 2023).

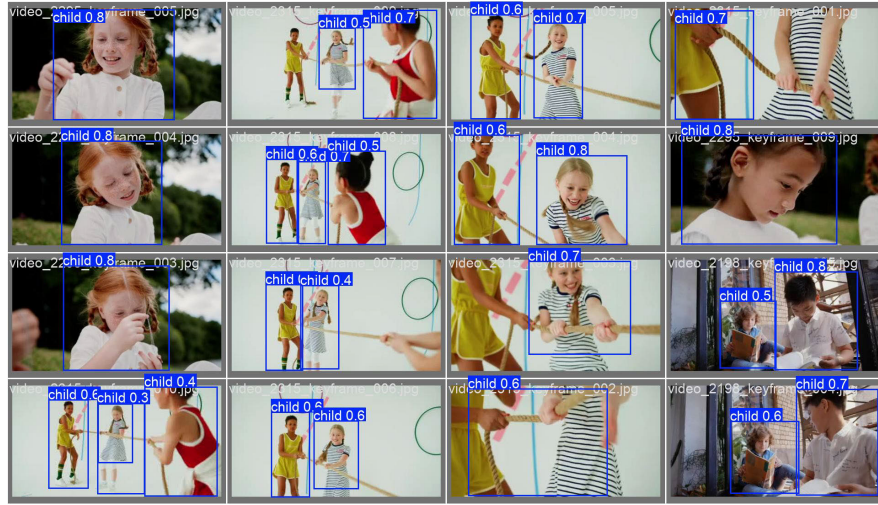


Figure 2.1: Sample results from the Child Detection Dataset (CDD) illustrating diverse pediatric postures and interactions.

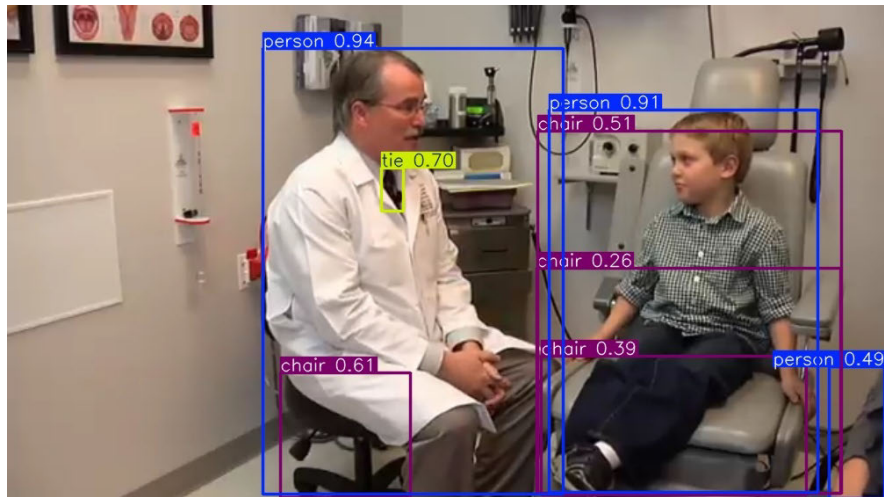


Figure 2.2: Comparative analysis: General YOLO limitations (left) versus specialized pediatric fine-tuning (right) in a clinical consultation scene.

CHAPTER 3

Mathematical Preliminaries

3.1 Object Detection & Instance Segmentation

The detection process is modeled as an optimization problem where we minimize a composite loss function $\mathcal{L}$ comprising box regression, class probability, and mask accuracy. Given an image I , the model outputs:

$$O = \{ (b_i, p_i, m_i) \}_{i=1}^k \quad (3.1)$$

Where b_i is the bounding box, p_i is the confidence score, and m_i is the pixel-level mask.

3.2 Allometric Scaling: The Biological Anchor

To distinguish between infants, adolescents, and adults, we utilize the scale-invariant Allometric Ratio R :

$$R = \frac{H_{total}}{H_{head}} \quad (3.2)$$

Where $R \approx 4.0 - 6.0$ for infants and $R \approx 7.5 - 8.0$ for adults [1].

3.3 Kalman Filtering for ID Persistence

To ensure unique patient counting and handle temporal occlusion, we utilize a linear Kalman Filter to predict the kinematic state x_k of a pediatric centroid (x, y) .

The filter operates in two phases:

1. Prediction Phase:

- State Prediction: $\hat{x}_k^- = F\hat{x}_{k-1} + Bu_k$
- Covariance Prediction: $P_k^- = FP_{k-1}F^T + Q$

2. Correction Phase:

- Kalman Gain: $K_k = P_k^- H^T (HP_k^- H^T + R)^{-1}$
- State Update: $\hat{x}_k = \hat{x}_k^- + K_k(z_k - H\hat{x}_k^-)$

Where F is the transition matrix and z_k is the measurement vector (detected centroid).

3.4 Performance Metric: MAPE

The system error is quantified using the Mean Absolute Percentage Error:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{Actual_i - Predicted_i}{Actual_i} \right| \times 100 \quad (3.3)$$

CHAPTER 4

Methodology

4.1 System Architecture

The proposed system follows a modular pipeline:

1. **Ingestion:** Low-res 640×480 clinical feed.
2. **Detection & Segmentation:** YOLOv11-seg identifies person masks.
3. **Validation (Ada-Haar):** A Haar Cascade classifier validates pediatric head signatures to ensure high confidence.
4. **Classification (Allometry):** Calculation of the R ratio for each candidate.
5. **Tracking (DeepSORT):** ID maintenance using Kalman Filters.

4.2 Handling Spatial Occlusion

When a single body bounds two distinct heads (h_1, h_2) , the system evaluates:

- Vertical Displacement: $y_{h2} < y_{h1}$.
- Area Proportion: $A_{h2} < A_{h1}$.

4.3 Edge Optimization

We utilize Region of Interest (RoI) pooling to prioritize pediatric candidates, maintaining 30+ FPS on edge hardware (Liu et al., 2020).

CHAPTER 5

Results & Discussion

5.1 System Performance Projections

Based on fine-tuning on the Child Detection Dataset (CDD), we project a Mean Absolute Percentage Error (MAPE) of less than 10% in controlled clinical environments. The integration of the Ada-Haar Booster provides a critical redundancy layer for low-resolution 640×480 feeds.

5.2 Discussion: Low-Res Robustness

The ability to maintain accuracy in low-light and low-resolution environments is the primary technical achievement. By using allometric ratios as a secondary verification, we bypass the “guessing” nature of deep learning in ambiguous scenes.

5.3 Ethical Considerations

The system operates exclusively on local edge hardware, ensuring that no patient video data is transmitted or stored beyond volatile memory. This satisfies the highest medical ethics standards for KATH.

CHAPTER 6

Conclusion

This research demonstrates that biological invariants, when synthesized with modern computer vision, can solve the long-standing challenge of the occluded pediatric census. Future work will focus on integrating these counts directly into the LHIMS dashboard for real-time staffing adjustments.

Bibliography

- [1] T. J. Mlinari et al. “Research on Children’s Body Proportions: Determining the Canon of Head Length to Total Body Height”. In: *Applied Sciences* 14.16 (2024), p. 7185.